Kernix 0.13.0 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

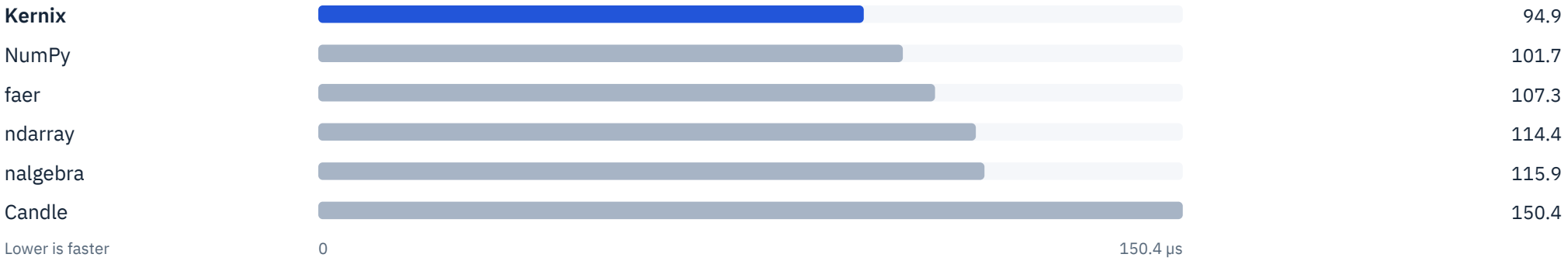
51 workloads

175 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.13.0 / Sparse matrices & memory

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse matrix × vector

Per-call median

2048 × 2048 · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



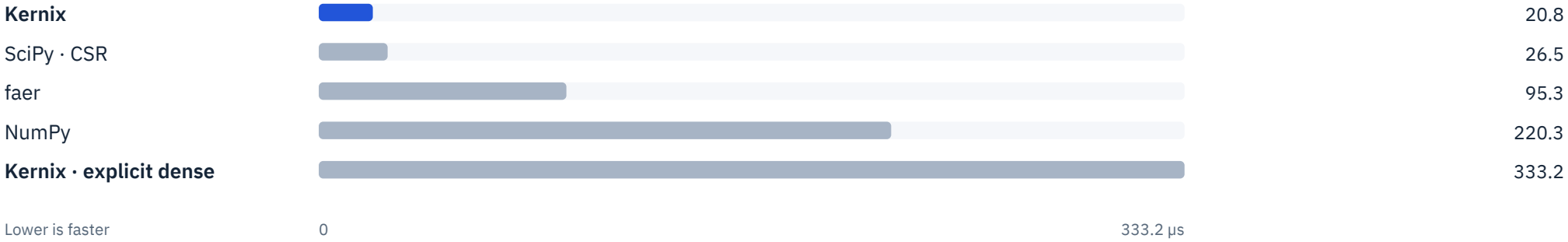
Kernix 0.13.0 / Sparse matrices & memory

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f32 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × sparse matrix

Per-call median

256 × 256 · five diagonals each · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Kernix 0.13.0 / Sparse matrices & memory

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Automatic CSR construction

Per-call median

512 × 512 · clone + scan + compress · f64 · Construction · dense copy or compression · preparation costs included



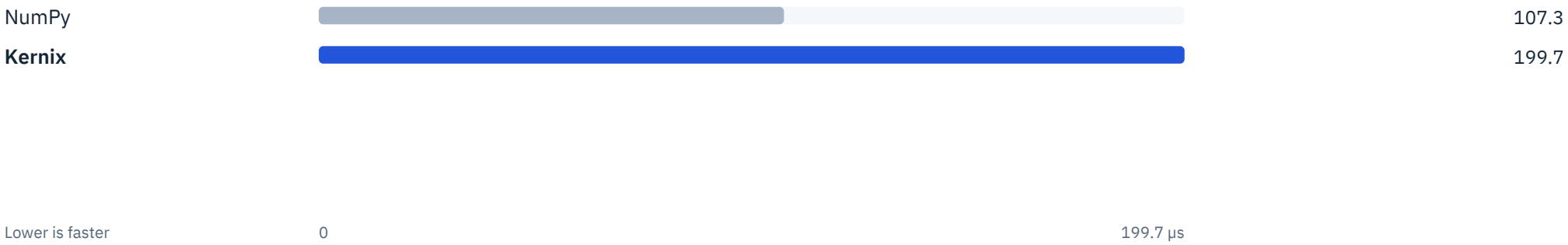
Kernix 0.13.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · transposed view

Per-call median

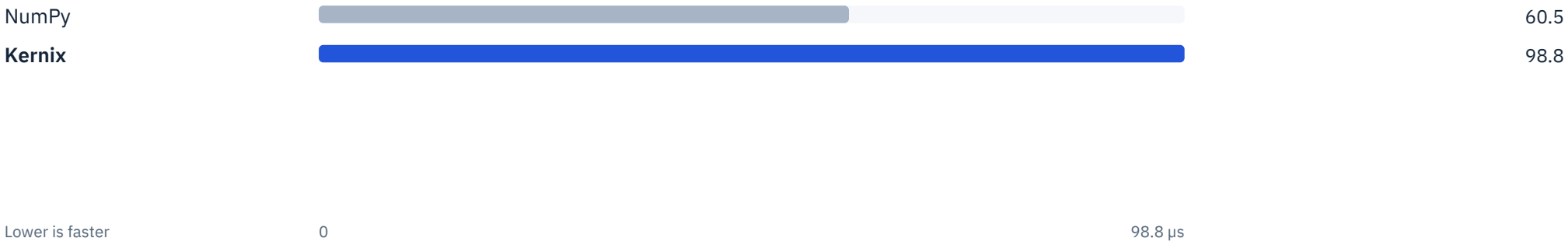
128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



Kernix 0.13.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



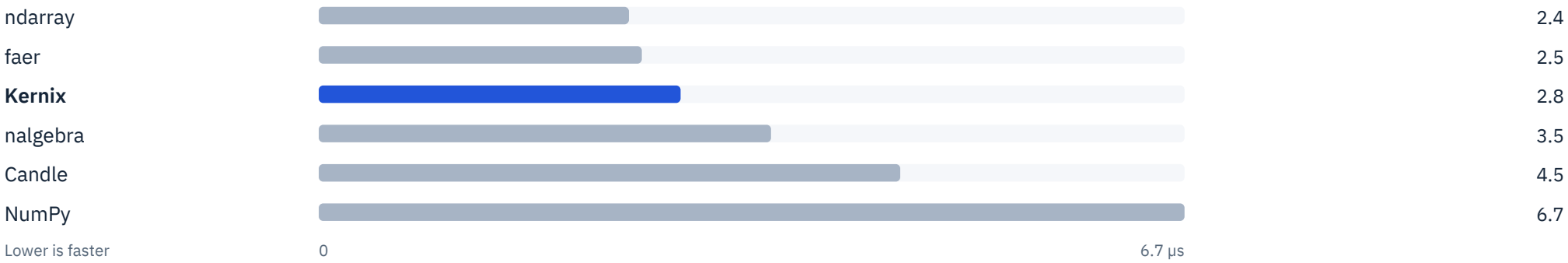
Kernix 0.13.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.13.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



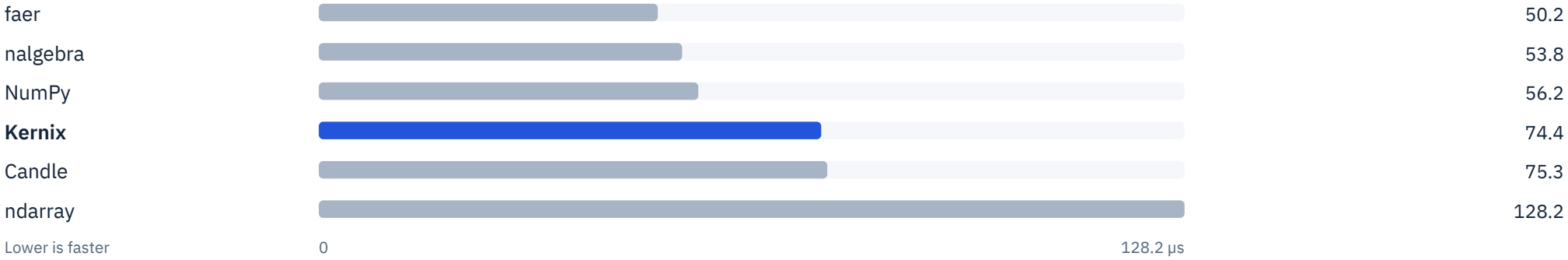
Kernix 0.13.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



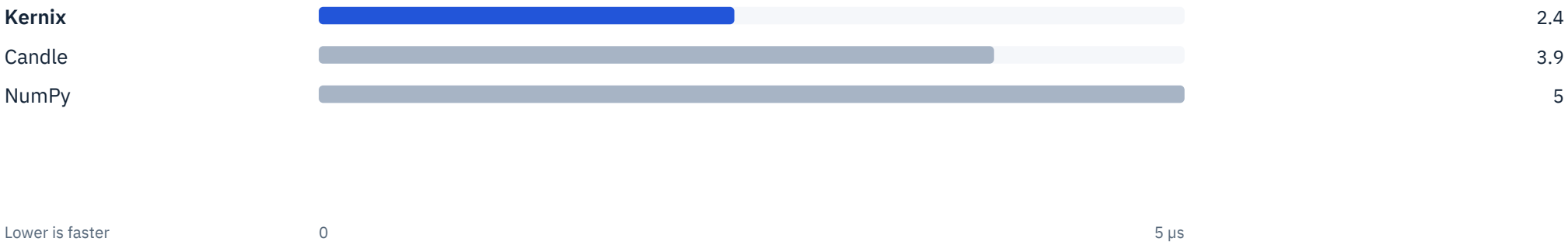
Kernix 0.13.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

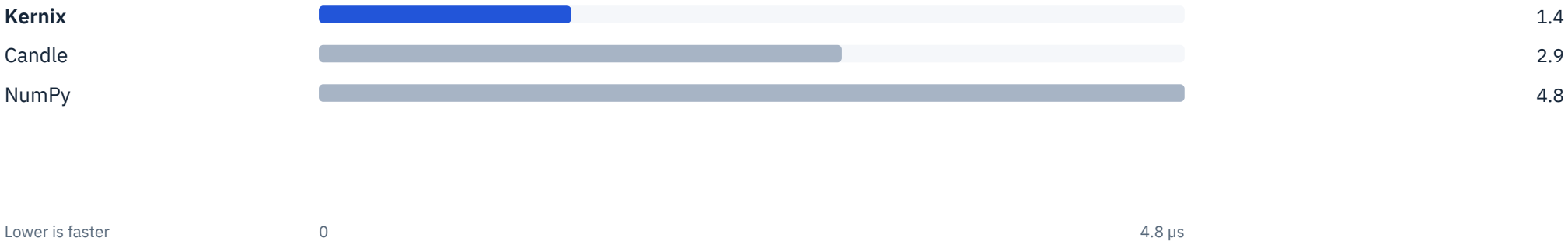
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.13.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.13.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

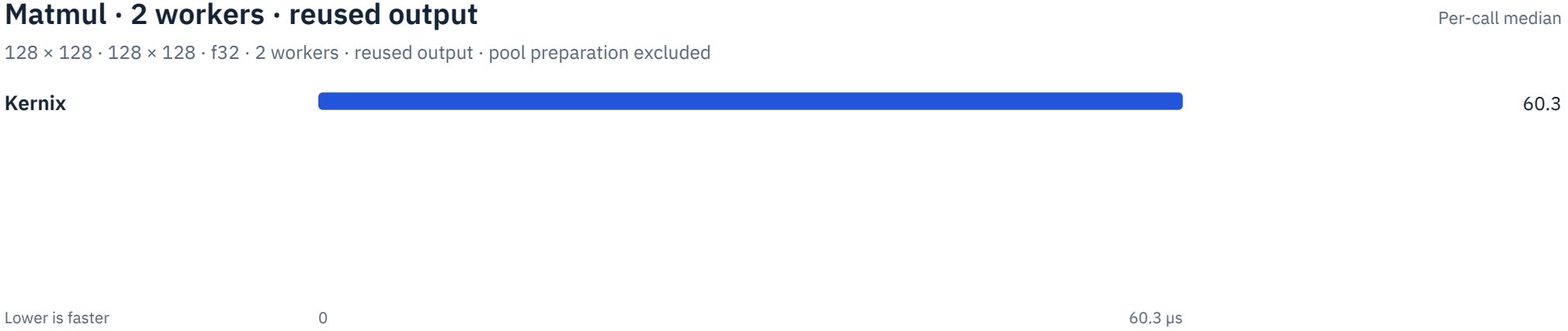
Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



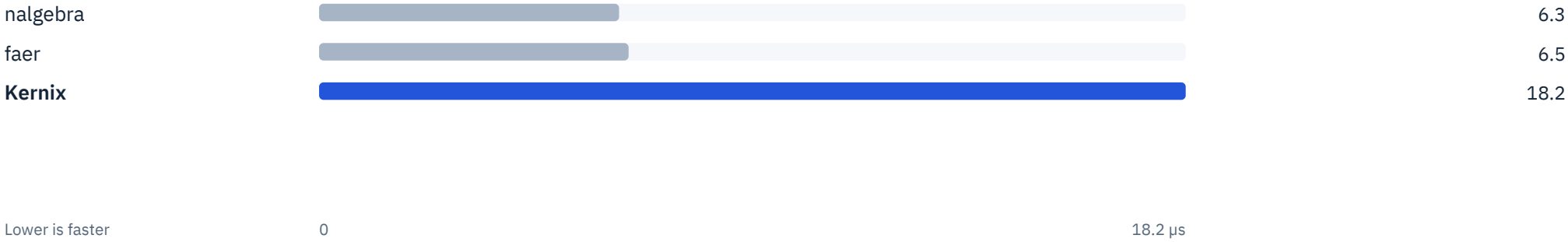
Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



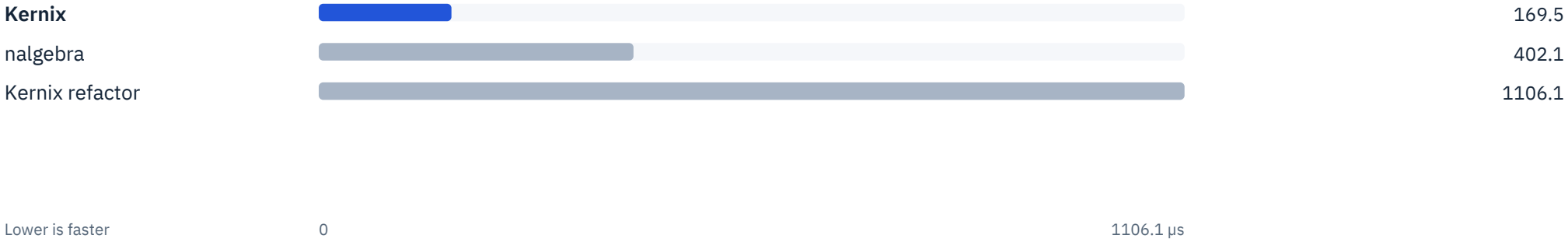
Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

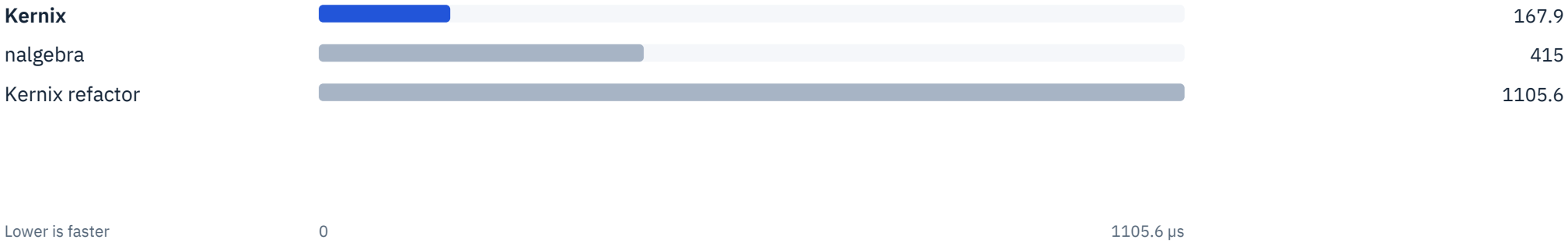
256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

Per-call median

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

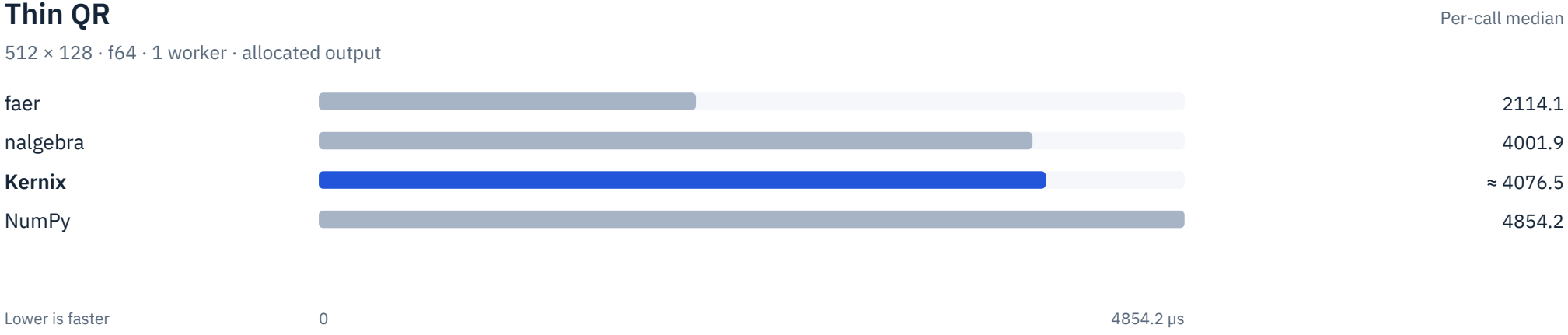
Cholesky rank-one update

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Thin QR

512 × 128 · f64 · 1 worker · allocated output

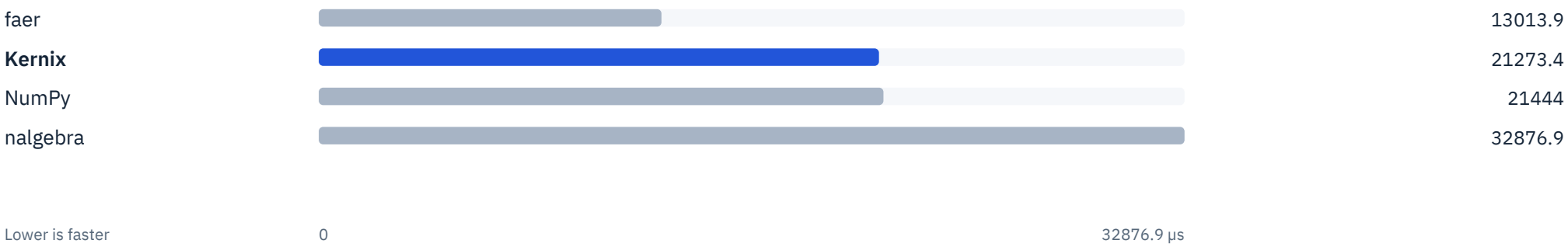


Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

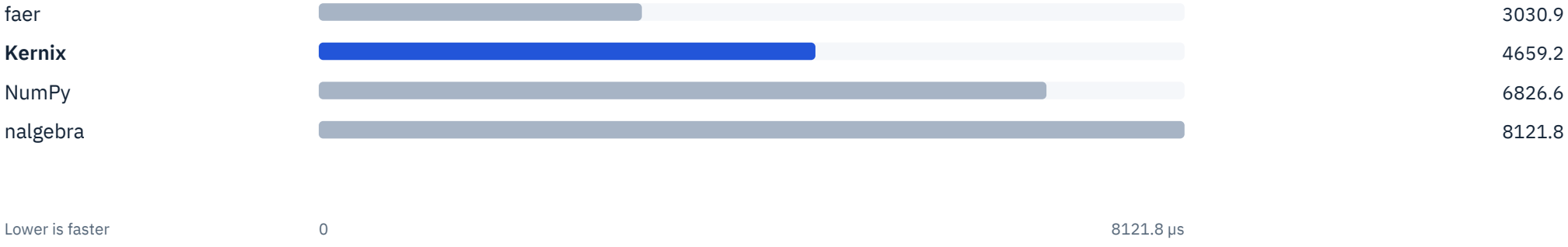
Thin QR

1024 × 256 · f64 · 1 worker · allocated output



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output



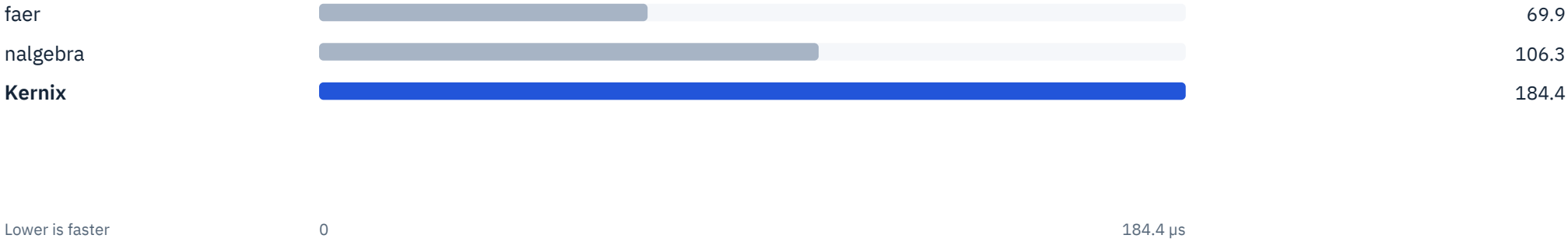
Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

Per-call median

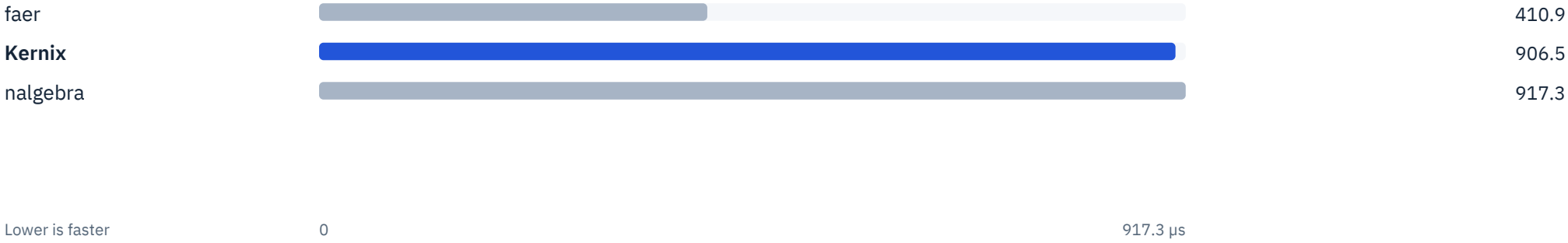
96 × 32 · f64 · 1 worker · allocated output



Column-pivoted QR

Per-call median

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

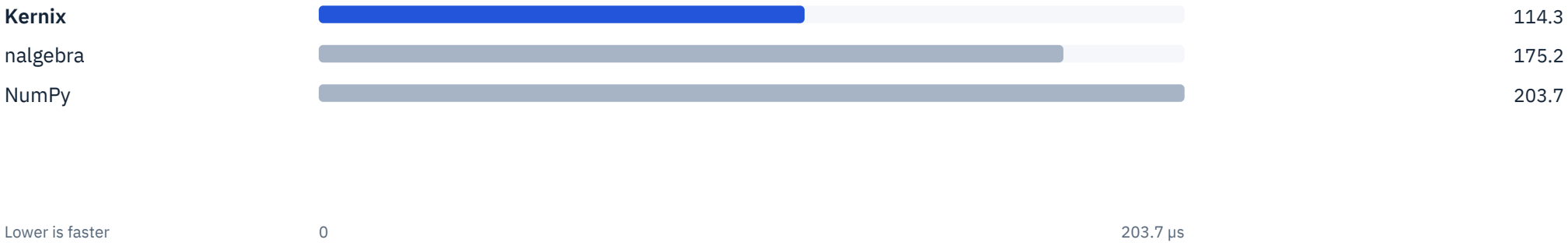
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



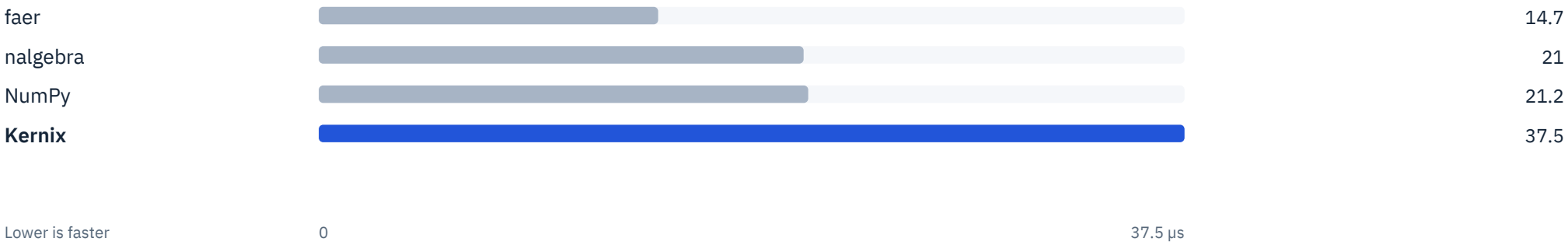
Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

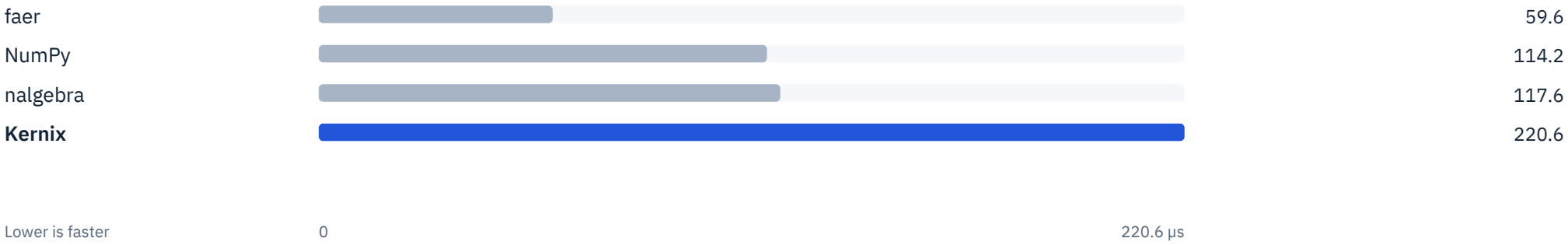
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.13.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.13.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.13.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

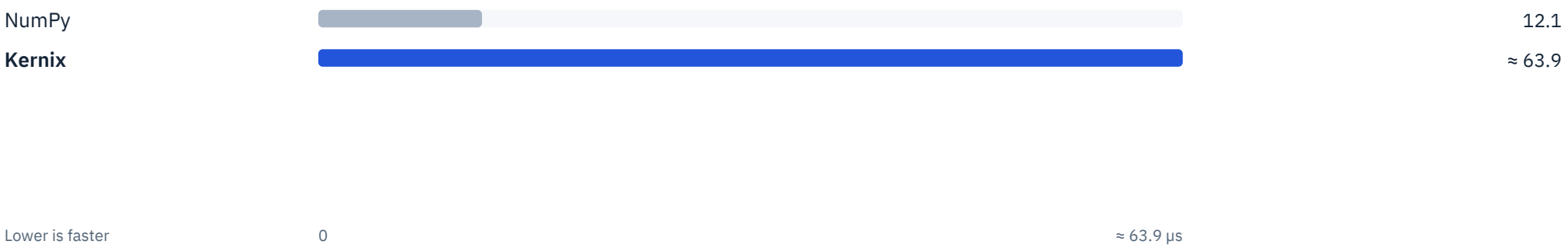


Kernix 0.13.0 / Tensor contractions

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

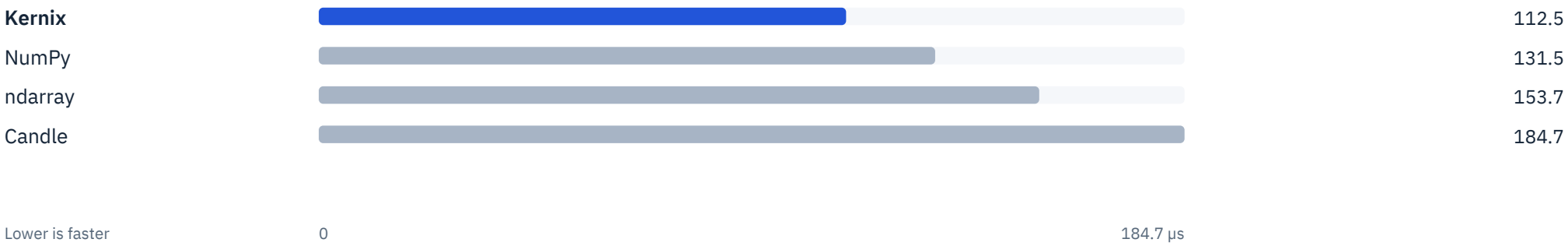


Kernix 0.13.0 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



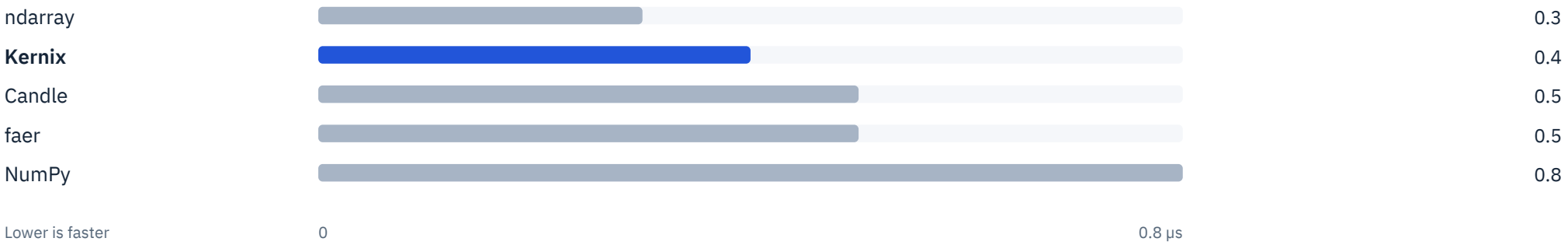
Kernix 0.13.0 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Timing direction unclear



CURRENT / BASELINE	1	95% INTERVAL	0.978494623655914 – 1

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Observed time increase



CURRENT / BASELINE	1.0877385048482953	95% INTERVAL	1.0835212691500116 – 1.0905533380292713

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.980936403843221	0.9784331656759641 – 0.9852986217457886

Thin QR

512 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9980635263938836	0.9794909809735605 – 1.004127710364034

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9930393565603598	0.9869423490620097 – 1.0008656185663485

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0135542168674698	0.5107446276861569 – 1.0339157245632067

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.020387359836901	0.990990990990991 – 1.0287769784172662

Thin QR

256 × 64 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0235950780159837	0.9791957376633261 – 1.0424158014687264

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1	1 – 1

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0010593220338984	0.9968354430379747 – 1.0031712473572938

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	1	95% INTERVAL	0.9957716701902748 – 1.0021231422505308

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	1.0054614964500272	95% INTERVAL	0.9927105831533477 – 1.0194106641721234

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9974789915966387	0.8314212128307188 – 1.0033670033670035

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.96	0.94 – 0.9795918367346939

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9978947368421053	0.9895324494068388 – 0.9992997198879552

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9922480620155039	0.9896507115135834 – 1.0025974025974025

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear



CURRENT / BASELINE	1	95% INTERVAL	0.9615384615384616 – 1.04

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear



CURRENT / BASELINE	1	95% INTERVAL	1 – 1

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9961055819991346	0.9587852494577006 – 1.0378208014407924

Cholesky factorization

256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9976703552708212	0.9940622737146995 – 1.0071553738317758

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.999373094965304	0.9815937955664504 – 1.0053508504807065

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0118906064209274	0.7700320512820513 – 1.256198347107438

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9625962596259626	0.6264762427904422 – 1.0512041150339022

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1	0.9807383627608346 – 1.25705329153605

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse matrix × vector / faer 2048 × 2048 · five diagonals · f64	7900.000000000001	7.900000000000001
Sparse matrix × vector / SciPy · CSR 2048 × 2048 · five diagonals · f64	11200	11.2
Sparse matrix × vector / NumPy 2048 × 2048 · five diagonals · f64	815400	815.4
Sparse matrix × vector / Kernix · explicit dense 2048 × 2048 · five diagonals · f64	2588100	2588.1
Sparse matrix × vector / Kernix 2048 × 2048 · five diagonals · f64	19300	19.3
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f64	42500	42.5
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f64	627900	627.9
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f64	469900	469.9
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f64	67500	67.5
Sparse × dense matrix / Kernix 512 × 512 · 32 RHS · five diagonals · f64	69200	69.2

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f32	26500	26.5
Sparse × dense matrix / Kernix 512 × 512 · 32 RHS · five diagonals · f32	20800	20.8
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f32	333200	333.2
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f32	220300	220.3
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f32	95300	95.3
Sparse × sparse matrix / NumPy 256 × 256 · five diagonals each · f64	759300	759.3
Sparse × sparse matrix / SciPy · CSR 256 × 256 · five diagonals each · f64	68300	68.3
Sparse × sparse matrix / Kernix · explicit dense 256 × 256 · five diagonals each · f64	756700	756.7
Sparse × sparse matrix / Kernix 256 × 256 · five diagonals each · f64	150300	150.3
Automatic CSR construction / Kernix 512 × 512 · clone + scan + compress · f64	1132200	1132.2

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Automatic CSR construction / SciPy · CSR 512 × 512 · clone + scan + compress · f64	1468900	1468.9
Automatic CSR construction / Kernix · explicit dense 512 × 512 · clone + scan + compress · f64	386400	386.4
Automatic CSR construction / NumPy 512 × 512 · clone + scan + compress · f64	414200	414.2
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	107300	107.3
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	199700	199.7
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	98800	98.8
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	60500	60.5
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	142100	142.1
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	101800	101.8
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	56200	56.2

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	74800	74.8
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2800	2.8
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2500	2.5
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	6700	6.7
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2400	2.4
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	4500	4.5
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	3500	3.5
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	60000	60
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	57900	57.9
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	13600	13.6

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14600	14.6
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	14200	14.2
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	16000	16
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	114400	114.4
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	101700	101.7
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	107300	107.3
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	94900	94.9
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	115900	115.9
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	150400	150.4
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	865000	865

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1181800	1181.8
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	786400	786.4
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1213600	1213.6
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	856000	856
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	905100	905.1
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	10400	10.4
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	6900	6.9
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	7000	7
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7400	7.4
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	30000	30

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	30100	30.1
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	53800	53.8
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	128200	128.2
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	75300	75.3
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	74400	74.4
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	50200	50.2
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	56200	56.2
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	390800	390.8
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	430400	430.4
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	398000	398

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	451800	451.8
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	388700	388.7
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	433300	433.3
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3900	3.9
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	5000	5
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2400	2.4
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	4800	4.8
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2900	2.9
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1400	1.4
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	41100	41.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	83900	83.9
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	607900	607.9
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	60300	60.3
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	18200	18.2
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	6500	6.5
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	6300	6.3
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	328900	328.9
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1395700	1395.7
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1326800	1326.8
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	2259900	2259.9

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	10418700	10418.7
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	10320400	10320.4
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	373900	373.9
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1302100	1302.1
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1177900	1177.9
Cholesky rank-one update / Kernix 256 × 256 · f64	169500	169.5
Cholesky rank-one update / nalgebra 256 × 256 · f64	402100	402.1
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	1106100	1106.1
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	415000	415
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	1105600	1105.6

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky rank-one downdate / Kernix 256 × 256 · f64	167900	167.9
Cholesky rank-one update / nalgebra 512 × 512 · f64	558800	558.8
Cholesky rank-one update / Kernix 512 × 512 · f64	1341000	1341
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	4640400	4640.4
Thin QR / NumPy 512 × 128 · f64	4854200	4854.2
Thin QR / Kernix 512 × 128 · f64	4076500.0000000005	4076.5000000000005
Thin QR / nalgebra 512 × 128 · f64	4001900	4001.9
Thin QR / faer 512 × 128 · f64	2114100	2114.1
Thin QR / nalgebra 1024 × 256 · f64	32876900	32876.9
Thin QR / faer 1024 × 256 · f64	13013900	13013.9

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / Kernix 1024 × 256 · f64	21273400	21273.4
Thin QR / NumPy 1024 × 256 · f64	21444000	21444
Cholesky factorization / Kernix 512 × 512 · f64	4659200	4659.2
Cholesky factorization / NumPy 512 × 512 · f64	6826600	6826.6
Cholesky factorization / nalgebra 512 × 512 · f64	8121800	8121.8
Cholesky factorization / faer 512 × 512 · f64	3030900	3030.9
Column-pivoted QR / nalgebra 96 × 32 · f64	106300	106.3
Column-pivoted QR / Kernix 96 × 32 · f64	184400	184.4
Column-pivoted QR / faer 96 × 32 · f64	69900	69.9
Column-pivoted QR / Kernix 256 × 64 · f64	906500	906.5

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / nalgebra 256 × 64 · f64	917300	917.3
Column-pivoted QR / faer 256 × 64 · f64	410900	410.9
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	103500	103.5
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	96800	96.8
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	112800	112.8
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	114300	114.3
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	175200	175.2
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	203700	203.7
Thin QR / Kernix 96 × 32 · f64	100200	100.2
Thin QR / nalgebra 96 × 32 · f64	79600	79.6

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 96 × 32 · f64	76700	76.7
Thin QR / faer 96 × 32 · f64	86000	86
Thin QR / faer 256 × 64 · f64	388200	388.2
Thin QR / Kernix 256 × 64 · f64	771500	771.5
Thin QR / NumPy 256 × 64 · f64	548000	548
Thin QR / nalgebra 256 × 64 · f64	580000	580
Cholesky factorization / faer 64 × 64 · f64	14700	14.7
Cholesky factorization / Kernix 64 × 64 · f64	37500	37.5
Cholesky factorization / nalgebra 64 × 64 · f64	21000	21
Cholesky factorization / NumPy 64 × 64 · f64	21200	21.2

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	220600	220.6
Cholesky factorization / faer 128 × 128 · f64	59600	59.6
Cholesky factorization / NumPy 128 × 128 · f64	114200	114.2
Cholesky factorization / nalgebra 128 × 128 · f64	117600	117.6
Cholesky factorization / nalgebra 256 × 256 · f64	853900	853.9
Cholesky factorization / faer 256 × 256 · f64	314100	314.1
Cholesky factorization / NumPy 256 × 256 · f64	947100	947.1
Cholesky factorization / Kernix 256 × 256 · f64	705700	705.7
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	58000	58
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	33100	33.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	4000	4
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	9900	9.9
Model inference / Candle batched normalized/sliced MLP · f64	8200	8.2
Model inference / Kernix batched normalized/sliced MLP · f64	3600	3.6
Model recording + forward / Candle batched normalized/sliced MLP · f64	9700	9.7
Model recording + forward / Kernix batched normalized/sliced MLP · f64	5000	5
Model backward / Kernix prepared graph → fresh gradients · f64	6000	6
Model backward / Candle prepared graph → fresh gradients · f64	32000	32
Parameter update / Kernix transactional SGD, fixed gradients · f64	1000	1
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	12100	12.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	63899.99999999999	63.89999999999999
Row softmax / ndarray 256 × 64 · f64	153700	153.7
Row softmax / NumPy 256 × 64 · f64	131500	131.5
Row softmax / Candle 256 × 64 · f64	184700	184.7
Row softmax / Kernix 256 × 64 · f64	112500	112.5
Create zeros / faer 64 × 64 · f32	500	0.5
Create zeros / Candle 64 × 64 · f32	500	0.5
Create zeros / ndarray 64 × 64 · f32	300	0.3
Create zeros / Kernix 64 × 64 · f32	400	0.4
Create zeros / NumPy 64 × 64 · f32	800	0.8

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create ones / faer 64 × 64 · f32	500	0.5
Create ones / ndarray 64 × 64 · f32	400	0.4
Create ones / Candle 64 × 64 · f32	700	0.7
Create ones / NumPy 64 × 64 · f32	2100	2.1
Create ones / Kernix 64 × 64 · f32	500	0.5